

# Enzymes I — Nature, Classification, Cofactors, Active Site

Syllabus lines 1–3

PART 1 / 6

## 01 DEFINITION, PROPERTIES & IUB CLASSIFICATION

**Enzymes** are biocatalysts — proteins (except **ribozymes**, catalytic RNA) elaborated by living cells that **accelerate a reaction by lowering its activation energy**, without being consumed and **without altering  $\Delta G$  or the equilibrium constant**. They only make equilibrium arrive sooner.

**Protein nature** — thermolabile, denatured by pH extremes, non-dialysable

**Efficiency**  $10^6$ – $10^{12}$  fold acceleration

**Active in trace amounts**

**Highly specific**

**Reaction reversible**, enzyme unchanged

**Regulatable** — allosteric, covalent, induction

| EC CLASS                         | REACTION CATALYSED                                                                             | SUB-GROUPS                                                  | VETERINARY EXAMPLE                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>1 • Oxidoreductases</b>       | Transfer of electrons / hydrogen (redox)                                                       | Dehydrogenases, oxidases, peroxidases, catalase, oxygenases | <b>LDH, SDH, cytochrome oxidase, glutathione peroxidase</b>                           |
| <b>2 • Transferases</b>          | Transfer of a group (other than H) between molecules                                           | Transaminases, kinases, methyl/acyl transferases            | <b>AST, ALT, hexokinase, GGT</b>                                                      |
| <b>3 • Hydrolases</b>            | Cleavage of a bond by addition of water                                                        | Esterases, glycosidases, peptidases, phosphatases           | <b>Lipase, amylase, trypsin, ALP, cholinesterase, urease</b>                          |
| <b>4 • Lyases</b>                | Removal of a group <b>non-hydrolytically</b> , leaving a double bond (or the reverse addition) | Decarboxylases, aldolases, hydratases, synthases            | <b>Aldolase, fumarase, carbonic anhydrase, HMG-CoA lyase</b>                          |
| <b>5 • Isomerases</b>            | Intramolecular rearrangement                                                                   | Racemases, epimerases, mutases                              | <b>Phosphohexose isomerase, phosphoglucomutase, methylmalonyl-CoA mutase</b>          |
| <b>6 • Ligases (synthetases)</b> | Joining of two molecules, <b>coupled to ATP hydrolysis</b>                                     | C–C, C–N, C–O, C–S ligases                                  | <b>Pyruvate carboxylase, glutamine synthetase, DNA ligase, acetyl-CoA carboxylase</b> |

**Six classes in order — "Over The Hill, Little Indian Lady":** Oxidoreductase • Transferase • Hydrolase • Lyase • Isomerase • Ligase. **EC number** has four digits: **class** . **sub-class** . **sub-sub-class** . **serial no.** — e.g. EC 1.1.1.27 = lactate dehydrogenase.

**Trap:** a **synthase** (lyase, class 4) needs **no** ATP; a **synthetase** (ligase, class 6) **must** use ATP. Similarly **oxidase** uses O<sub>2</sub> as acceptor, **oxygenase** incorporates O<sub>2</sub> into the product, **dehydrogenase** uses NAD/FAD and never O<sub>2</sub>.

## 02 COFACTOR · COENZYME · PROSTHETIC GROUP

Apoenzyme protein, inactive + Cofactor metal or coenzyme = Holoenzyme active

**Coenzyme** **Loosely, non-covalently bound**, dissociates freely, behaves as a **second substrate / co-substrate** and must be regenerated — NAD<sup>+</sup>, NADP<sup>+</sup>, CoA, THF.

**Prosthetic group** **Tightly or covalently bound**, stays with the enzyme through the cycle — FAD, biotin, haem, pyridoxal phosphate, lipoic acid.

**Activator** Metal ion required for activity or for substrate binding (metal-activated enzymes); if firmly held the enzyme is a **metalloenzyme**.

| VITAMIN                           | COENZYME                             | REACTION TYPE                                                      | DEFICIENCY IN ANIMALS                                                                                   |
|-----------------------------------|--------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>B<sub>1</sub> Thiamine</b>     | TPP                                  | <b>Oxidative decarboxylation</b> , transketolase                   | <b>Polioencephalomalacia</b> (thiaminase, ruminants); <b>Chastek paralysis</b> — raw fish, fox/mink/cat |
| <b>B<sub>2</sub> Riboflavin</b>   | FMN, FAD                             | Hydrogen transfer                                                  | Curled-toe paralysis in chicks                                                                          |
| <b>B<sub>3</sub> Niacin</b>       | NAD <sup>+</sup> , NADP <sup>+</sup> | Hydride transfer                                                   | <b>Black tongue</b> in dogs; from tryptophan 60 : 1                                                     |
| <b>B<sub>5</sub> Pantothenate</b> | CoA, ACP                             | Acyl transfer                                                      | Goose-stepping in pigs, dermatitis                                                                      |
| <b>B<sub>6</sub> Pyridoxine</b>   | PLP                                  | <b>Transamination, decarboxylation</b> , deamination, ALA synthase | Convulsions, microcytic anaemia                                                                         |
| <b>B<sub>7</sub> Biotin</b>       | Biocytin                             | <b>Carboxylation (CO<sub>2</sub> fixation)</b>                     | Bound by <b>avidin</b> of raw egg white; foot-pad lesions in poultry                                    |
| <b>B<sub>9</sub> Folate</b>       | THF                                  | <b>One-carbon transfer</b>                                         | Megaloblastic anaemia; sulphonamide target in bacteria                                                  |
| <b>B<sub>12</sub> Cobalamin</b>   | Methyl- & deoxyadenosyl-cobalamin    | <b>Methylmalonyl-CoA mutase</b> , homocysteine methylation         | <b>Cobalt deficiency in ruminants</b> — blocks propionate → glucose; "pine/wasting disease"             |
| <b>Lipoic acid</b>                | Lipoamide                            | Acyl transfer in PDH / α-KGDH                                      | —                                                                                                       |

Zn carbonic anhydrase, carboxypeptidase, LDH

Mg all kinases & ATP reactions

Mn arginase

Cu cytochrome oxidase, tyrosinase, SOD, ceruloplasmin

Fe catalase, peroxidase, cytochromes

Se glutathione peroxidase — **white muscle disease**

Mo xanthine oxidase

Ni urease

### 03 ISOENZYMES (ISOZYMES)

**Definition:** physically distinct forms of the **same** enzyme that catalyse the **same** reaction but differ in amino-acid sequence, **electrophoretic mobility**,  $K_m$ , **heat stability**, **immunological properties and tissue distribution**. They arise from different genes or different subunit combinations.

| ENZYME | STRUCTURE                                                                                                   | DIAGNOSTIC VALUE                                                                                                                                |
|--------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| LDH    | <b>Tetramer of H and M subunits</b><br>→ <b>5 isoenzymes</b> ( $H_4$ , $H_3M$ , $H_2M_2$ , $HM_3$ , $M_4$ ) | <b>LDH-1 (<math>H_4</math>)</b> heart & RBC · <b>LDH-5 (<math>M_4</math>)</b> liver & skeletal muscle. Haemolysis falsely raises total LDH      |
| CK     | Dimer of M and B                                                                                            | <b>CK-MM</b> skeletal muscle · <b>CK-MB</b> myocardium · <b>CK-BB</b> brain. Best marker of <b>acute muscle damage</b> (short half-life)        |
| ALP    | Separate gene loci                                                                                          | Liver, bone (young growing animals — physiologically high), intestinal, placental and — <b>unique to the dog — corticosteroid-induced C-ALP</b> |
| AST    | Cytosolic + mitochondrial                                                                                   | Mitochondrial form appears only in severe necrosis                                                                                              |

**Why it matters:** total enzyme activity says **something is damaged**; the isoenzyme pattern says **which organ**. This is the direct bridge to Unit 3 clinical enzymology.

### 04 ENZYME-SUBSTRATE COMPLEX, THE MODERN ACTIVE CENTRE & SPECIFICITY



**Two models of the active site:**

**Lock & key**  
(Emil Fischer, 1894)

Active site is a **rigid, pre-formed template** exactly complementary to the substrate. Explains **specificity** well, but not the action of enzymes on several related substrates.

**Induced fit**  
(Koshland, 1958)

**The modern concept.** The active site is **flexible**; the approaching substrate **induces a conformational change** that brings the catalytic groups into correct alignment — like a hand entering a glove. Explains allosteric regulation, competitive inhibition and non-productive binding.

**Features of the active centre:** ① a small region — only **3–5 % of the total enzyme volume**; ② a **crevice or cleft** from which water is largely excluded; ③ built from amino-acid residues that are **far apart in the primary sequence but brought together by folding**; ④ has a **substrate-binding (contact) site** and a **catalytic site**; ⑤ binding is by weak, non-covalent forces — H-bond, ionic, hydrophobic, van der Waals; ⑥ residues commonly involved: **serine, histidine, cysteine, aspartate, glutamate, lysine, tyrosine** (the **Ser-His-Asp catalytic triad** of trypsin and chymotrypsin); ⑦ the shape complements the **transition state**, not the ground-state substrate.

| TYPE OF SPECIFICITY         | MEANING                                                     | EXAMPLE                                                                                                                                                                             |
|-----------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Absolute</b>             | Acts on <b>one substrate only</b>                           | <b>Urease</b> → <b>urea</b> ; carbonic anhydrase; lactase                                                                                                                           |
| <b>Substrate (relative)</b> | Acts on a group of closely related substrates               | Alcohol dehydrogenase on several alcohols; hexokinase on glucose, fructose, mannose                                                                                                 |
| <b>Group</b>                | Acts on a particular chemical <b>group</b> next to the bond | <b>Trypsin</b> — carboxyl side of <b>lysine &amp; arginine</b> ; <b>chymotrypsin</b> — aromatic residues; <b>pepsin</b> — aromatic                                                  |
| <b>Bond</b>                 | Acts on a bond type regardless of the rest                  | Esterases (ester), glycosidases (glycosidic), peptidases (peptide), lipases                                                                                                         |
| <b>Stereo / optical</b>     | Acts on only <b>one optical or geometric isomer</b>         | <b>L-amino acid oxidase</b> vs D-amino acid oxidase; $\alpha$ - vs $\beta$ -glycosidase; <b>fumarase acts on trans-fumarate but not cis-maleate</b> ; only <b>D-sugars</b> are used |
| <b>Dual / co-substrate</b>  | Two substrates required                                     | Transaminases (amino acid + keto acid); dehydrogenases (substrate + NAD <sup>+</sup> )                                                                                              |

**Most-asked one-liner: "Stereospecificity of fumarase"** — it hydrates fumarate (trans) but never maleate (cis). This one example answers both "optical specificity" and "why enzymes are more selective than chemical catalysts".

**Trap: zymogens / proenzymes** are catalytically inactive precursors activated by **limited proteolysis**: pepsinogen → pepsin (HCl, then autocatalysis), **trypsinogen → trypsin by enteropeptidase**, chymotrypsinogen, procarboxypeptidase, prothrombin, plasminogen. Premature activation inside the pancreas = **acute pancreatitis**.

**05 FACTORS INFLUENCING ENZYME ACTION**

① **Substrate concentration.** At low  $[S]$  the rate is **first order** ( $\propto [S]$ ); as  $[S]$  rises the curve flattens to a **rectangular hyperbola** and becomes **zero order** at saturation, when every active site is occupied  $\rightarrow V_{\max}$ .

**Michaelis–Menten**  $v = V_{\max}[S] / (K_m + [S])$

$K_m$  = the substrate concentration at which  $v = \frac{1}{2}V_{\max}$  (units of concentration, mol/L).

- **Low  $K_m$  = high affinity** for the substrate (hexokinase  $K_m$  0.1 mM) and vice versa (glucokinase 10 mM).
- $K_m$  is a **constant characteristic of the enzyme–substrate pair** and is **independent of enzyme concentration**;  $V_{\max}$  is **directly proportional** to it.
- **Lineweaver–Burk (double reciprocal) plot** linearises the curve:  $1/v = (K_m/V_{\max})(1/[S]) + 1/V_{\max}$  — y-intercept  $1/V_{\max}$ , x-intercept  $-1/K_m$ , slope  $K_m/V_{\max}$ .

② **Enzyme concentration.** With substrate in excess the velocity is **directly proportional to  $[E]$**  — a straight line through the origin. This linearity is the basis of every enzyme assay in the clinical laboratory.

③ **Temperature.** Rate roughly **doubles for each 10 °C rise ( $Q_{10} \approx 2$ )** up to the **optimum ( $\approx 37\text{--}40\text{ °C}$  in mammals,  $41\text{--}42\text{ °C}$  in birds)**, beyond which thermal denaturation causes a sharp fall. The curve is therefore **bell-shaped, but asymmetrical**. Enzymes are inactivated above  $\sim 60\text{ °C}$ ; hence samples are refrigerated.

④ **pH.** Bell-shaped curve; extremes alter the ionisation of the active site and of the substrate, and finally denature the enzyme.

Pepsin 1.5–2.0

Acid phosphatase 5

Salivary amylase 6.8

Trypsin / lipase 8

ALP 9–10

Arginase 9.5

⑤ **Others:** time of incubation, accumulation of product (product inhibition), activators (metal ions, bile salts for lipase), inhibitors, light and UV/ionising radiation.

**Enzyme units — learn all four definitions verbatim.**

**International Unit (IU)** The amount of enzyme that converts **1  $\mu\text{mol}$  of substrate per minute** under defined optimum conditions of temperature, pH and substrate concentration.

**Katal (SI unit)** The amount of enzyme that converts **1 mole of substrate per second**.  $1\text{ katal} = 6 \times 10^7\text{ IU}$ ;  $1\text{ IU} = 16.67\text{ nkat}$ .

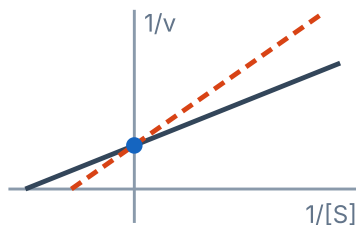
|                                                             |                                                                                                                                                                                                                            |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Specific activity</b>                                    | <b>Units of enzyme activity per milligram of protein</b> — the index of <b>purity</b> ; it rises at every purification step.                                                                                               |
| <b>Turnover number</b><br>( $k_{cat}$ , molecular activity) | The <b>number of substrate molecules converted per molecule of enzyme (or per active site) per unit time when the enzyme is fully saturated.</b><br>Highest known: <b>carbonic anhydrase</b> $\approx$ 600 000 per second. |

**Trap:** specific activity measures **purity**; turnover number measures **catalytic efficiency** of a single molecule. Students routinely swap the two — a guaranteed one-mark loss.

## 06 ENZYME INHIBITION — THE FOUR TYPES, DECIDED BY THE LINEWEAVER-BURK PLOT

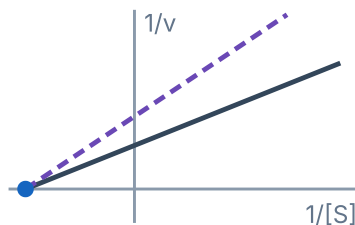
Solid line = no inhibitor · dashed line = with inhibitor · blue dot = the point both lines share.

### COMPETITIVE



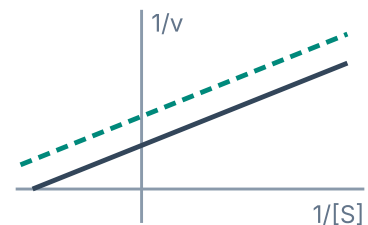
Lines meet on the **y-axis** (same  $1/V_{max}$ ).  $K_m \uparrow$ ,  $V_{max}$  unchanged.

### NON-COMPETITIVE



Lines meet on the **x-axis** (same  $-1/K_m$ ).  $V_{max} \downarrow$ ,  $K_m$  unchanged.

### UNCOMPETITIVE



**Parallel lines** (same slope). Both  $K_m$  and  $V_{max} \downarrow$ .

| TYPE                   | BINDS TO                                                                                      | $K_M$        | $V_{MAX}$    | REVERSED BY $\uparrow[S]$ ? | CLASSIC EXAMPLES                                                                                                                                                                                            |
|------------------------|-----------------------------------------------------------------------------------------------|--------------|--------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Competitive</b>     | <b>Active site</b> of free enzyme; inhibitor is a <b>structural analogue</b> of the substrate | $\uparrow$   | Unchanged    | <b>Yes</b>                  | <b>Malonate on succinate dehydrogenase</b> · sulphonamides vs PABA · allopurinol on xanthine oxidase · ethanol on alcohol dehydrogenase in <b>ethylene-glycol poisoning</b> · neostigmine on cholinesterase |
| <b>Non-competitive</b> | A site <b>other than</b> the active site, on E or ES; unrelated structurally                  | Unchanged    | $\downarrow$ | <b>No</b>                   | Heavy metals ( $Pb^{2+}$ , $Hg^{2+}$ , $Ag^+$ ) on $-SH$ groups · <b>cyanide &amp; azide on cytochrome oxidase</b> · fluoride on enolase · iodoacetate on GAPDH                                             |
| <b>Uncompetitive</b>   | <b>Only the ES complex</b> , never the free enzyme                                            | $\downarrow$ | $\downarrow$ | No                          | Phenylalanine on intestinal ALP; lithium on inositol monophosphatase                                                                                                                                        |

|                                                 |                                                                                                |                                               |    |                                                                                                                                                                                                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Irreversible / suicide (mechanism-based)</b> | Covalent, permanent — the enzyme <b>itself converts the inhibitor</b> into the reactive killer | Activity lost until new enzyme is synthesised | No | <b>Organophosphates &amp; carbamates on acetylcholinesterase</b> (the commonest small-animal poisoning; antidote atropine + 2-PAM) · aspirin on COX · penicillin on transpeptidase · allopurinol → alloxanthine · 5-fluorouracil on thymidylate synthase |
|-------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**The two answers examiners want in full:** ① **malonate vs succinate dehydrogenase** — malonate differs from succinate by one CH<sub>2</sub> group, occupies the active site, raises apparent K<sub>m</sub>, and inhibition is relieved by excess succinate. ② **Organophosphate poisoning** — irreversible phosphorylation of the serine of acetylcholinesterase, "ageing" of the enzyme, and the rationale of oxime therapy **before** ageing occurs.

## 07 ALLOSTERIC ENZYMES & THE FOUR MODES OF METABOLIC REGULATION

**Allosteric ("other site") enzymes** are the regulatory enzymes of a pathway. They are **oligomeric**, possess a **catalytic site plus a separate allosteric site**, and bind their effectors non-covalently and reversibly.

- They give a **sigmoid (S-shaped) v vs [S] curve, not a hyperbola** — because subunits show **co-operativity**; they therefore **do not obey Michaelis–Menten kinetics**.
- **Positive effectors (activators)** shift the curve left (↑affinity); **negative effectors (inhibitors)** shift it right.
- **Homotropic** = the substrate is itself the effector (O<sub>2</sub> on haemoglobin); **heterotropic** = a different molecule (ATP on PFK-1).
- Models: **MWC concerted / symmetry model** (all subunits flip T ⇌ R together) and **KNF sequential model** (one at a time).
- Almost always the **first committed, irreversible, rate-limiting step** of the pathway, and usually subject to **feedback (end-product) inhibition**.

**Learn these five allosteric enzymes with their effectors:** **PFK-1** (– ATP, citrate; + AMP, fructose-2,6-bisphosphate) · **pyruvate carboxylase** (+ acetyl-CoA) · **glycogen phosphorylase** (+ AMP in muscle; – ATP, G6P) · **carbamoyl phosphate synthetase I** (+ N-acetylglutamate) · **HMG-CoA reductase** and **acetyl-CoA carboxylase** (+ citrate, – palmitoyl-CoA).

| MODE OF REGULATION                      | SPEED                 | MECHANISM & EXAMPLE                                                                                                                                                                                                      |
|-----------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Allosteric modulation</b>            | <b>Seconds</b>        | Non-covalent binding of an effector; feedback inhibition.<br><b>PFK-1, ATCase</b>                                                                                                                                        |
| <b>Covalent modification</b>            | Seconds–minutes       | <b>Phosphorylation / dephosphorylation</b> by hormone-driven kinases & phosphatases. <b>Glycogen phosphorylase ON, glycogen synthase OFF when phosphorylated</b> (glucagon/adrenaline); insulin does the reverse         |
| <b>Proteolytic (zymogen) activation</b> | Minutes, irreversible | Digestive enzymes, clotting cascade, complement                                                                                                                                                                          |
| <b>Induction / repression</b>           | <b>Hours–days</b>     | Change in the <b>amount</b> of enzyme by altered transcription. <b>Glucokinase and lipogenic enzymes induced by insulin; gluconeogenic enzymes by cortisol &amp; glucagon; cytochrome P450 induced by phenobarbitone</b> |
| <b>Compartmentation</b>                 | —                     | Opposing pathways kept apart: <b>β-oxidation in the mitochondrion, fatty-acid synthesis in the cytosol</b>                                                                                                               |

**Trap:** phosphorylation does **not** mean "activation". It activates glycogen phosphorylase and hormone-sensitive lipase but **inactivates** glycogen synthase, acetyl-CoA carboxylase and pyruvate dehydrogenase. Remember it as "**phosphorylation switches the body to the fasting mode**".

**08 ENZYMES & COENZYMES OF OXIDATION-REDUCTION**

**Biological oxidation** = removal of electrons (usually as hydrogen atoms) from a substrate, coupled to reduction of an acceptor. It proceeds in **small steps down a redox-potential gradient** so the energy is captured, not lost as a flash of heat.

| ENZYME CLASS                      | WHAT IT DOES                                                                                                                          | EXAMPLE                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Dehydrogenases</b>             | Transfer H to <b>NAD<sup>+</sup>/NADP<sup>+</sup></b> or <b>FAD/FMN</b> , never to O <sub>2</sub>                                     | LDH, MDH, isocitrate DH, succinate DH                                                                                   |
| <b>Oxidases</b>                   | Use <b>O<sub>2</sub> as the direct acceptor</b> → H <sub>2</sub> O or H <sub>2</sub> O <sub>2</sub>                                   | <b>Cytochrome oxidase</b> (→H <sub>2</sub> O); xanthine oxidase, D-amino acid oxidase (→H <sub>2</sub> O <sub>2</sub> ) |
| <b>Peroxidases &amp; catalase</b> | <b>Dispose of H<sub>2</sub>O<sub>2</sub></b> ; catalase: 2H <sub>2</sub> O <sub>2</sub> → 2H <sub>2</sub> O + O <sub>2</sub>          | Glutathione peroxidase ( <b>Se</b> ), myeloperoxidase, catalase ( <b>Fe</b> )                                           |
| <b>Oxygenases</b>                 | <b>Dioxygenase</b> — both O atoms into substrate; <b>Monoxygenase / mixed-function oxidase</b> — one O into substrate, one into water | <b>Cytochrome P450</b> (steroidogenesis, drug metabolism); homogentisate dioxygenase                                    |

|                                           |                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NAD<sup>+</sup> / NADP<sup>+</sup></b> | From <b>niacin</b> ; accept a <b>hydride ion (2e<sup>-</sup> + H<sup>+</sup>)</b> . <b>NADH → ETC for ATP</b> ; <b>NADPH → reductive biosynthesis</b> and antioxidant defence.                                          |
| <b>FAD / FMN</b>                          | From <b>riboflavin</b> ; accept <b>2H (2e<sup>-</sup> + 2H<sup>+</sup>)</b> ; tightly bound prosthetic groups.                                                                                                          |
| <b>Coenzyme Q (ubiquinone)</b>            | <b>Lipid-soluble, freely mobile</b> in the membrane; the <b>collection point</b> for electrons from Complex I, Complex II, ETF of β-oxidation and glycerol-3-phosphate dehydrogenase. Not protein, not vitamin-derived. |
| <b>Cytochromes</b>                        | Haemoproteins; transfer <b>one electron at a time</b> by Fe <sup>3+</sup> ⇌ Fe <sup>2+</sup> . Order b → C <sub>1</sub> → C → a → a <sub>3</sub> . <b>Cytochrome c is the only water-soluble, mobile one.</b>           |
| <b>Iron-sulphur proteins</b>              | Non-haem iron (Fe-S centres) in Complexes I, II, III; one-electron carriers.                                                                                                                                            |

**-7.3**

ATP, kcal/mol

**-14.8**

PEP — highest

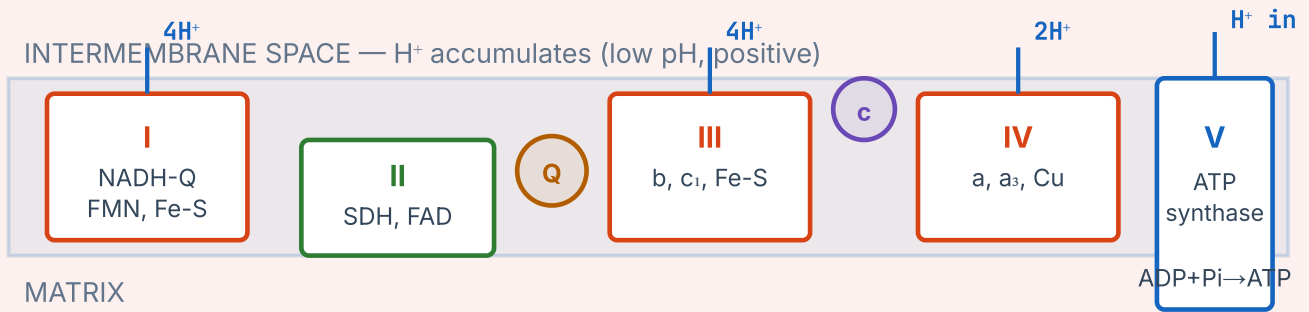
**-11.8**

1,3-BPG

**-10.3**

creatine-P

## 09 THE RESPIRATORY CHAIN — FOUR COMPLEXES



$NADH \rightarrow FMN \rightarrow CoQ \rightarrow cyt\ b \rightarrow c_1 \rightarrow c \rightarrow a \rightarrow a_3 \rightarrow \frac{1}{2}O_2 + 2H^+ \rightarrow H_2O$

**Rotenone/amytal**  $\perp$  **I** • **Malonate/carboxin**  $\perp$  **II** • **Antimycin A/BAL**  $\perp$  **III**

**Cyanide, CO, azide,  $H_2S$**   $\perp$  **IV** • **Oligomycin**  $\perp$  **V** • **Attractyloside**  $\perp$  **ADP/ATP translocase**

Electrons always flow from the **more negative to the more positive redox potential**;

Complex II pumps **no** protons — that is why  $FADH_2$  yields less ATP than NADH.

## 10 OXIDATIVE PHOSPHORYLATION, UNCOUPLERS & THE SHUTTLES

**Chemiosmotic theory — Peter Mitchell, 1961 (Nobel 1978).** Electron flow through Complexes I, III and IV pumps  $H^+$  from the matrix into the intermembrane space, creating a **proton-motive force** made of a pH gradient ( $\Delta pH$ ) plus a membrane potential ( $\Delta \psi$ ). Protons re-enter **only** through the  $F_0$  channel of **ATP synthase (Complex V)**, and this flow drives **rotational catalysis** in  $F_1$  (Boyer's binding-change mechanism) which releases ATP.

- **Prerequisite: an intact, impermeable inner mitochondrial membrane.** Any leak abolishes ATP synthesis while electron flow continues.
- **Respiratory control:** the rate of the chain is governed by the availability of **ADP** — the cell makes ATP only as fast as it spends it.
- **P : O ratio** = molecules of ATP formed per atom of oxygen reduced: **NADH = 2.5 (classically 3)**,  **$FADH_2$  = 1.5 (classically 2)**.

**Uncouplers** carry  $H^+$  back across the membrane, short-circuiting the gradient. **Electron transport continues (even accelerates) but ATP is not made — the energy appears as heat.**

Signs: hyperthermia, tachypnoea, weight loss, no ATP gain.

2,4-Dinitrophenol, dinitro-o-cresol

CCCP, FCCP

Salicylate in overdose

Unconjugated bilirubin

Excess thyroxine, free fatty acids

Thermogenin (UCP-1) — physiological

**Veterinary gold: brown adipose tissue** of the newborn lamb, piglet, kid and kitten is packed with **UCP-1/thermogenin**, a **deliberate, hormone-controlled uncoupler** that generates heat instead of ATP — **non-shivering thermogenesis**. Its failure or absence explains neonatal hypothermia and much of the "weak-neonate" complex.

**Trap:** distinguish sharply — an **inhibitor** stops electron flow **and** ATP synthesis and oxygen consumption falls; an **uncoupler** lets electron flow and O<sub>2</sub> consumption **increase** while ATP falls; **oligomycin** blocks the ATP synthase itself, so both stop — and adding an uncoupler **restores** respiration after oligomycin. That last sentence is a favourite viva question.

**Shuttles.** The inner membrane is impermeable to NADH, so cytosolic NADH from glycolysis must be transferred indirectly — and the choice of shuttle decides the final ATP count for glucose.

| SHUTTLE                     | DELIVERS AS                 | ATP        | TISSUE                                       |
|-----------------------------|-----------------------------|------------|----------------------------------------------|
| <b>Glycerol-3-phosphate</b> | FADH <sub>2</sub> (via CoQ) | <b>1.5</b> | Skeletal muscle, brain, insect flight muscle |
| <b>Malate-aspartate</b>     | NADH                        | <b>2.5</b> | Liver, heart, kidney — more efficient        |

**32**

ATP/glucose, malate-aspartate

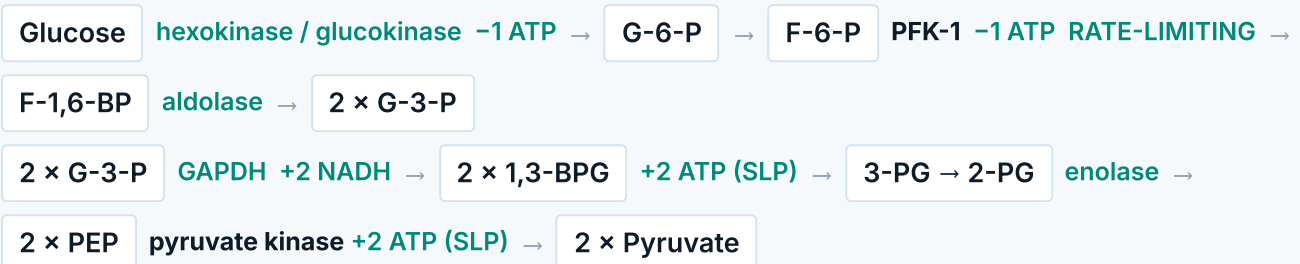
**30**

ATP/glucose, glycerol-P

**38 / 36**

classical figures

**Other transporters of the inner membrane:** adenine-nucleotide translocase (ADP in / ATP out — blocked by **atractyloside**), phosphate translocase, tricarboxylate (citrate) carrier, and the **carnitine shuttle** for fatty acids.

**11 GLYCOLYSIS (EMBDEN-MEYERHOF PATHWAY) — CYTOSOL, EVERY CELL, AEROBIC OR ANAEROBIC**


- **Three irreversible, regulatory steps:** hexokinase/glucokinase, **PFK-1 (the key rate-limiting enzyme)**, pyruvate kinase. These are exactly the three bypassed in gluconeogenesis.
- **Balance sheet:** 2 ATP invested, 4 ATP generated by **substrate-level phosphorylation** → **net 2 ATP + 2 NADH per glucose**. Anaerobically the 2 NADH are spent reducing pyruvate to **lactate (LDH)** to regenerate  $\text{NAD}^+$ , so the net stays 2 ATP — the only ATP source for the **RBC (no mitochondria)**, working muscle, cornea, renal medulla and rapidly growing tissue.
- **Hexokinase vs glucokinase:** hexokinase is in all tissues,  $K_m$  0.1 mM, **inhibited by G-6-P**; **glucokinase** is in liver and  $\beta$ -cell,  $K_m \approx 10$  mM, **not** inhibited by G-6-P and **induced by insulin** — it only works after a meal, letting the liver take up glucose **only when blood glucose is high**.
- **Rapoport-Luebering shunt** in the RBC diverts 1,3-BPG to **2,3-BPG**, which lowers haemoglobin's  $\text{O}_2$  affinity and improves tissue delivery — at the cost of one ATP.

**Practical link you will be asked:** **sodium fluoride inhibits enolase**, so NaF-containing (grey-top) tubes are used for blood glucose collection — without it, erythrocyte glycolysis destroys ~10 % of the glucose per hour and gives a falsely low result. Also: **iodoacetate** blocks GAPDH, **arsenate** uncouples at GAPDH (glycolysis runs but makes no ATP).

**Link reaction — pyruvate dehydrogenase complex** (mitochondrial matrix):  $\text{pyruvate} + \text{CoA} + \text{NAD}^+ \rightarrow \text{acetyl-CoA} + \text{CO}_2 + \text{NADH}$ . A multienzyme complex of 3 enzymes needing **5 coenzymes: TPP, Lipoamide, CoA, FAD,  $\text{NAD}^+$** . **Irreversible** — this single fact is **why fatty acids can never be converted to glucose** in animals. Inhibited by **arsenite** and by mercury (bind lipoamide -SH); switched off by phosphorylation (PDH kinase, activated by ATP/NADH/acetyl-CoA) and on by dephosphorylation (PDH phosphatase,  $\text{Ca}^{2+}$ , insulin).

**PDH coenzymes — "Tender Loving Care For Nancy":** TPP, Lipoamide, CoA, FAD,  $\text{NAD}^+$ . The  $\alpha$ -ketoglutarate dehydrogenase complex uses exactly the same five.

## 12 KREBS / TCA / CITRIC ACID CYCLE

**Site:** mitochondrial matrix (succinate dehydrogenase alone is in the inner membrane). **Nature:** the **final common oxidative pathway** for carbohydrate, fat and protein, and **amphibolic** — it both degrades and supplies precursors. Strictly **aerobic** (needs  $\text{NAD}^+$ /FAD regenerated by the ETC), though  $\text{O}_2$  takes no direct part.

| STEP                                         | ENZYME                               | YIELD / NOTE                                                                            |
|----------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Acetyl-CoA + OAA</b>                      | Citrate synthase                     | Condensation; the entry point                                                           |
| <b>Citrate → isocitrate</b>                  | Aconitase                            | <b>Blocked by fluoroacetate</b> ("1080", <b>Dichapetalum</b> /Gifblaar plant poisoning) |
| <b>Isocitrate → <math>\alpha</math>-KG</b>   | <b>Isocitrate DH — rate limiting</b> | 1st NADH + 1st $\text{CO}_2$                                                            |
| <b><math>\alpha</math>-KG → succinyl-CoA</b> | $\alpha$ -KG dehydrogenase complex   | 2nd NADH + 2nd $\text{CO}_2$ ; same 5 coenzymes as PDH                                  |
| <b>Succinyl-CoA → succinate</b>              | Succinyl-CoA synthetase              | <b>GTP by substrate-level phosphorylation</b> — the only one in the cycle               |
| <b>Succinate → fumarate</b>                  | Succinate DH (Complex II)            | <b><math>\text{FADH}_2</math></b> ; membrane bound; <b>inhibited by malonate</b>        |
| <b>Fumarate → malate</b>                     | Fumarase                             | Stereospecific hydration                                                                |
| <b>Malate → OAA</b>                          | Malate DH                            | 3rd NADH; OAA regenerated                                                               |

3

NADH

1

$\text{FADH}_2$

1

GTP

10

ATP per acetyl-CoA

2

$\text{CO}_2$  released

- **Anaplerotic ("topping-up") reactions** replace intermediates drained for biosynthesis: **pyruvate carboxylase** (pyruvate → OAA, biotin, activated by acetyl-CoA) — the chief one; transamination of glutamate/aspartate; and in the ruminant, **propionate → succinyl-CoA**.
- **Regulation:** by the **ATP/ADP and NADH/ $\text{NAD}^+$  ratios** at citrate synthase, isocitrate DH and  $\alpha$ -KG DH;  $\text{Ca}^{2+}$  activates all three during muscular work.
- **Amphibolic outputs:** citrate → fatty acid synthesis;  $\alpha$ -KG & OAA → amino acids & gluconeogenesis; succinyl-CoA → **haem synthesis**; fumarate ← urea cycle.

## 13 HMP SHUNT (PENTOSE PHOSPHATE PATHWAY)

**Site:** cytosol of **liver, adipose tissue, lactating mammary gland, adrenal cortex, gonads, RBC and leucocytes** — all tissues that need NADPH. Also called the Warburg–Dickens–Horecker or hexose monophosphate pathway. **It produces no ATP and consumes none.**

G-6-P → G-6-P dehydrogenase · **rate-limiting** · **NADPH #1** → 6-phosphogluconate

6-PG DH · **NADPH #2 +  $\text{CO}_2$**  → Ribulose-5-P

Ribulose-5-P → Ribose-5-P **non-oxidative, reversible: transketolase (TPP) & transaldolase** →

F-6-P + G-3-P back to glycolysis

- **Two products, both irreplaceable: NADPH** (2 per G-6-P; 12 per glucose fully cycled) and **ribose-5-phosphate** for nucleotides, ATP, NAD, FAD, CoA.
- **What NADPH is spent on:** ① **fatty acid, cholesterol and steroid-hormone synthesis** — hence the pathway's intense activity in adipose tissue, mammary gland and adrenal cortex; ② **reduction of glutathione (glutathione reductase) → GSH**, the RBC's shield against oxidant damage; ③ the **respiratory burst** of neutrophils via NADPH oxidase; ④ **cytochrome P450** hydroxylations in detoxification; ⑤ reduction of folate and of methaemoglobin.

**Clinical anchor:** when the NADPH–glutathione shield is overwhelmed, haemoglobin is oxidised → **Heinz-body haemolytic anaemia and methaemoglobinaemia**. Classic Indian-practice causes: **onion and Brassica/kale poisoning in cattle and dogs, paracetamol in the cat, red maple in the horse, and phenothiazine**.

#### 14 GLUCONEOGENESIS · CORI CYCLE · GLYCOGEN METABOLISM · BIOENERGETICS

**Gluconeogenesis** — synthesis of glucose from non-carbohydrate sources. **Site: liver (≈90 %) and renal cortex (≈10 %)**; spans cytosol, mitochondrion and endoplasmic reticulum.

**Substrates:** lactate, glycerol, **propionate (the ruminant's chief one)**, and glucogenic amino acids — **alanine above all**.

| BYPASS ENZYME                      | REACTION & NOTE                                                                                                         |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Pyruvate carboxylase</b>        | Pyruvate → OAA; <b>mitochondrial, biotin, needs acetyl-CoA as obligatory activator</b>                                  |
| <b>PEP carboxykinase</b>           | OAA → PEP; uses <b>GTP</b> ; induced by cortisol & glucagon                                                             |
| <b>Fructose-1,6-bisphosphatase</b> | <b>Rate-limiting</b> ; inhibited by AMP and by fructose-2,6-bisphosphate                                                |
| <b>Glucose-6-phosphatase</b>       | In the ER of <b>liver and kidney only</b> — muscle lacks it, which is why muscle glycogen can never raise blood glucose |

**Cost:** 6 high-energy phosphates (4 ATP + 2 GTP) plus 2 NADH per glucose — gluconeogenesis is expensive, which is why the liver will not run it and glycolysis at the same time (reciprocal regulation).

**Golden rule:** **acetyl-CoA (and therefore even-chain fatty acids) can never be converted to glucose** — the PDH step is irreversible. Only glycerol and odd-chain/propionate carbons from fat are glucogenic.

**Cori cycle (lactic acid cycle).** Muscle glycolysis under anaerobic work → lactate → blood → liver → gluconeogenesis → glucose → back to muscle. It **shifts the metabolic burden of the muscle to the liver** and clears lactic acid, at a net cost of **4 ATP to the liver** for every 2 recovered by muscle.

Muscle glucose → lactate → blood → Liver lactate → glucose → blood → muscle

The parallel **glucose–alanine (Cahill) cycle** carries both the carbon **and** the amino nitrogen of muscle protein to the liver, where alanine is transaminated to pyruvate for gluconeogenesis and its nitrogen enters the urea cycle.

**Glycogenesis.** Glucose → G-6-P → G-1-P (phosphoglucomutase) → UDP-glucose (uses UTP) → glycogen. **Glycogen synthase** makes the α1→4 chain on the primer protein **glycogenin**; **branching enzyme** (amylo-4,6-transglucosidase) transfers blocks of ~7 residues to make α1→6 branches. Activated by **insulin** (dephosphorylation).

**Glycogenolysis.** **Glycogen phosphorylase** (rate-limiting, needs **PLP**) carries out **phosphorolysis**, releasing **G-1-P**; the **debranching enzyme** (transferase + α1,6-glucosidase) releases the ~8 % of residues at branch points as free glucose. Driven by **glucagon and adrenaline via cAMP → protein kinase A**, which simultaneously switches synthase off — **one signal, opposite effects, no futile cycle**.

**Ruminant special (worth a whole long answer):** almost no glucose is absorbed from the gut, so the ruminant runs **gluconeogenesis continuously, all day**, mainly from **propionate**: propionate → propionyl-CoA → methylmalonyl-CoA → (B<sub>12</sub>, mutase) → succinyl-CoA → OAA → glucose. **Cobalt/vitamin B<sub>12</sub> deficiency blocks this step** and causes wasting; and any surge in demand (peak lactation, twin pregnancy) that outruns propionate supply produces **ketosis and pregnancy toxaemia** — the whole of Unit 3's energy section rests on this paragraph.

| PROCESS                                      | ATP                            |
|----------------------------------------------|--------------------------------|
| Glycolysis, anaerobic                        | +2                             |
| Glycolysis, aerobic (to pyruvate)            | +5 or +7                       |
| PDH ×2                                       | +5                             |
| TCA ×2 acetyl-CoA                            | +20                            |
| Glucose → CO <sub>2</sub> + H <sub>2</sub> O | 32 (30 via glycerol-P shuttle) |
| Gluconeogenesis, cost                        | -6                             |
| Glycogen → lactate (from G-1-P)              | +3                             |

**Trap: glycogen storage diseases** — **von Gierke** (type I, glucose-6-phosphatase; severe fasting hypoglycaemia + hepatomegaly), **Pompe** (type II, lysosomal α-glucosidase; reported in cattle and cats), **McArdle** (type V, muscle phosphorylase; exercise intolerance with **no** rise in blood lactate).

**15  $\beta$ -OXIDATION OF FATTY ACIDS (KNOOP)**

 Fatty acid  
cytosol

 acyl-CoA synthetase (thiokinase) ·  $\text{ATP} \rightarrow \text{AMP} + \text{PPi} = 2 \sim \text{P}$  spent  $\rightarrow$  Acyl-CoA

 carnitine shuttle: CPT-I (rate-limiting, inhibited by malonyl-CoA)  $\rightarrow$  translocase  $\rightarrow$  CPT-II  $\rightarrow$  Matrix

**Four repeating steps, each cycle shortening the chain by two carbons — mnemonic "Ox-Hy-Ox-Th":**

1. **Oxidation** — acyl-CoA dehydrogenase  $\rightarrow$   $\text{FADH}_2$
2. **Hydration** — enoyl-CoA hydratase
3. **Oxidation** —  $\beta$ -hydroxyacyl-CoA dehydrogenase  $\rightarrow$   $\text{NADH}$
4. **Thiolysis** — thiolase  $\rightarrow$  **acetyl-CoA** + acyl-CoA (n-2)

**Palmitate (C16) bioenergetics — memorise the arithmetic: 7 cycles  $\rightarrow$  8 acetyl-CoA + 7  $\text{FADH}_2$  + 7  $\text{NADH}$ .**
 $(8 \times 10) + (7 \times 1.5) + (7 \times 2.5) = 80 + 10.5 + 17.5 = 108 - 2 \text{ (activation)} = 106$   
 ATP (classical figures:  $129 - 2 = 129$ ).

 Cycles =  $(n/2) - 1$ ; acetyl-CoA =  $n/2$ .

- **Carnitine** is synthesised from **lysine + methionine** and is the obligatory ferry for long-chain fatty acids; short and medium chains enter freely. **Malonyl-CoA inhibits CPT-I** — the elegant switch that stops the cell burning and building fat at once.
- **Odd-chain fatty acids** end in **propionyl-CoA  $\rightarrow$  methylmalonyl-CoA  $\rightarrow$  (B<sub>12</sub>)  $\rightarrow$  succinyl-CoA** — the only part of a fatty acid that is **glucogenic**. Unsaturated fatty acids need an extra **isomerase and reductase**. **Very-long-chain fatty acids are shortened in peroxisomes**;  $\alpha$ -oxidation handles branched phytanic acid;  $\omega$ -oxidation is a minor microsomal route.

**16 KETONE-BODY FORMATION & FATTY ACID BIOSYNTHESIS**
**Ketogenesis — liver mitochondria only, by the HMG-CoA pathway.**

 2 Acetyl-CoA  $\rightarrow$  Acetoacetyl-CoA  $\xrightarrow{\text{HMG-CoA synthase — rate-limiting}}$  HMG-CoA

 HMG-CoA lyase  $\rightarrow$  Acetoacetate

 Acetoacetate  $\xrightarrow{\text{BHB dehydrogenase, NADH}}$   **$\beta$ -hydroxybutyrate — the major one**

 spontaneous decarboxylation  $\rightarrow$  **Acetone — exhaled, "sweet breath"**

- **The liver makes ketones but cannot use them** — it lacks **thiophorase (succinyl-CoA : acetoacetate CoA transferase)**. Heart, skeletal muscle, renal cortex and, in prolonged starvation, the **brain** oxidise them; ketone bodies are water-soluble, need no carrier and **spare glucose**.

- **Trigger in every species:** low insulin / high glucagon → unrestrained lipolysis → acetyl-CoA floods the liver, while **oxaloacetate is drained away into gluconeogenesis**, so acetyl-CoA cannot enter the TCA cycle. Hence the classical statement — "**fat burns in the flame of carbohydrate.**"
- **Causes:** starvation, high-fat/low-carbohydrate diet, **diabetes mellitus**, **bovine ketosis** of early lactation, **pregnancy toxaemia** of the ewe and doe.

**Ruminant twist:**  $\beta$ -hydroxybutyrate is a **normal** ruminant metabolite — the rumen epithelium converts absorbed **butyrate** to BHB during every meal. Ketosis is therefore a matter of degree, and BHB is the field test: **<1.0 mmol/L normal · 1.2–1.4 subclinical · >2.6–3.0 clinical ketosis.**

**Fatty acid biosynthesis (de novo lipogenesis) — cytosol.** Acetyl-CoA is exported from the mitochondrion as **citrate** (ATP-citrate lyase). **Acetyl-CoA carboxylase is the rate-limiting enzyme:** it needs **biotin, CO<sub>2</sub> and ATP**, is **activated by citrate and insulin** and **inhibited by palmitoyl-CoA, glucagon and adrenaline**, and it makes **malonyl-CoA**. **Fatty acid synthase**, a multienzyme dimer carrying **ACP (phosphopantetheine, from pantothenic acid)**, then repeats condensation → reduction → dehydration → reduction and **stops at palmitate (C16)**. Per palmitate: **8 acetyl-CoA, 7 ATP, 14 NADPH** (from the HMP shunt and malic enzyme). Elongation and desaturation ( $\Delta 9$ ,  $\Delta 6$ ,  $\Delta 5$  only) follow in the ER.

**Species trap:** in **ruminants the principal site of lipogenesis is adipose tissue and the mammary gland, not the liver**, and the starting material is **acetate absorbed from the rumen**, not citrate-derived acetyl-CoA. This is why a low-fibre ration (less acetate) causes **milk-fat depression**.

## 17 PROTEIN METABOLISM — BIOSYNTHESIS, DEGRADATION AND THE FATE OF THE AMINO GROUP

**Amino acid pool & nitrogen balance.** Dietary protein, tissue protein turnover and de-novo synthesis feed a common pool ( $\approx 100$  g in an adult) drawn on for protein synthesis, hormones, purines, haem, creatine, and for energy. **Positive balance** = growth, pregnancy, lactation, convalescence; **negative balance** = starvation, fever, trauma, parasitism, protein-deficient ration.

**Degradation** follows two routes: the **ATP-dependent ubiquitin–proteasome system** in the cytosol for short-lived and abnormal proteins (tagged by polyubiquitin), and **lysosomal cathepsins** for extracellular and membrane proteins. **Biosynthesis** is ribosomal translation of mRNA (see Sheet 6).

**Transamination** Reversible transfer of an  $\alpha$ -amino group to an  $\alpha$ -keto acid. Coenzyme **pyridoxal phosphate (B<sub>6</sub>)**. **ALT:** alanine +  $\alpha$ -KG  $\rightleftharpoons$  pyruvate + glutamate. **AST:** aspartate +  $\alpha$ -KG  $\rightleftharpoons$  OAA + glutamate. **Every amino acid except lysine, threonine, proline and hydroxyproline** undergoes it. Its purpose is to **funnel all nitrogen into glutamate**. Leakage of ALT/AST into serum is the basis of liver and muscle profiling in Unit 3.

**Deamination**      **Oxidative** — **glutamate dehydrogenase** (mitochondrial, NAD(P)<sup>+</sup>, reversible, allosterically inhibited by ATP/GTP and activated by ADP/GDP) is the chief route and the only one that liberates ammonia rapidly; L- and D-amino acid oxidases (FMN/FAD) are minor and give H<sub>2</sub>O<sub>2</sub>. **Non-oxidative** — serine/threonine dehydratase, cysteine desulphydrase, histidase.

**Transdeamination**      **Transamination + oxidative deamination coupled** — the combined route by which the nitrogen of almost all amino acids finally reaches ammonia.

**Decarboxylation** (PLP-dependent) removes -COOH as CO<sub>2</sub> and yields the **biogenic amines** — small marks, frequently asked:

| AMINO ACID         | AMINE                                        | ACTION                             |
|--------------------|----------------------------------------------|------------------------------------|
| Histidine          | Histamine                                    | Allergy, gastric HCl, vasodilation |
| Tryptophan         | Serotonin → melatonin                        | Neurotransmitter, gut motility     |
| Tyrosine           | Dopamine → NA → adrenaline; DOPA → melanin   | Catecholamines, pigment            |
| Glutamate          | GABA                                         | Chief inhibitory transmitter       |
| Cysteine           | Taurine                                      | Bile salts; essential in the cat   |
| Ornithine / lysine | Putrescine, spermine, spermidine, cadaverine | Polyamines — cell proliferation    |
| Serine             | Ethanolamine → choline                       | Phospholipid, acetylcholine        |

Amines are inactivated by **monoamine and diamine oxidases**. In the gut, bacterial decarboxylation of unabsorbed amino acids yields putrefactive amines — relevant to Unit 3 digestive disorders.

**Ammonia transport.** Free NH<sub>3</sub> is intensely neurotoxic (it drains α-ketoglutarate from the brain's TCA cycle and swells astrocytes as glutamine accumulates). It travels in blood in two safe forms: **glutamine** (glutamine synthetase, ATP; released by glutaminase in liver and kidney) — the principal carrier — and **alanine** from muscle by the glucose-alanine cycle.

**Urea cycle (Krebs-Henseleit, 1932) — liver only; steps 1–2 mitochondrial, 3–5 cytosolic.**

| # | ENZYME                                  | PRODUCT / NOTE                                                                                                              |
|---|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Carbamoyl phosphate synthetase I</b> | NH <sub>3</sub> + CO <sub>2</sub> + 2 ATP; <b>rate-limiting</b> ; absolutely requires <b>N-acetylglutamate</b> as activator |
| 2 | Ornithine transcarbamoylase             | + ornithine → <b>citrulline</b>                                                                                             |
| 3 | Argininosuccinate synthetase            | + <b>aspartate</b> (the 2nd nitrogen); ATP → AMP + PPi = 2 ~P                                                               |

|   |                                                     |                                                                            |
|---|-----------------------------------------------------|----------------------------------------------------------------------------|
| 4 | Argininosuccinate lyase                             | → arginine + <b>fumarate</b> — the link to the TCA cycle ("Krebs bicycle") |
| 5 | <b>Arginase</b> (Mn <sup>2+</sup> , liver-specific) | → <b>UREA</b> + ornithine (recycled)                                       |

**Remember:** Ordinarily, Careless Crappers Are Also Frivolous About Urination — Ornithine, Carbamoyl-P, Citrulline, Argininosuccinate, Arginine, Fumarate, Urea.

**Cost: 4 ATP equivalents (3 ATP molecules) per urea.** One nitrogen from free ammonia, one from **aspartate**; the carbon from CO<sub>2</sub>.

**Clinical payoffs:** ① **Portosystemic shunt** in the dog — high blood ammonia, **low BUN**, ammonium biurate crystals, hepatic encephalopathy. ② **Urea poisoning in ruminants** — rumen urease liberates NH<sub>3</sub> faster than the liver can clear it; rumen pH rises, absorption accelerates; treat with **cold water + 5 % acetic acid (vinegar)**. ③ **Nitrogen recycling** — urea returns to the rumen in saliva and across the wall, where microbes reuse it, so ruminants thrive on non-protein nitrogen. ④ Ureotelic (mammals) vs **uricotelic (birds, reptiles)** vs ammonotelic (fish).

# Nucleotide Metabolism · DNA & RNA Synthesis · Integration

Plus the final recall bank

PART 6 / 6

## 18 PURINE & PYRIMIDINE METABOLISM

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purine —<br/>de novo</b>       | Built <b>on the ribose</b> : R-5-P → (PRPP synthetase) → <b>PRPP</b> → (glutamine-PRPP amidotransferase — <b>committed, rate-limiting step, feedback-inhibited by AMP, GMP and IMP</b> ) → 11 steps → <b>IMP</b> → AMP (needs <b>GTP</b> + aspartate) or GMP (needs <b>ATP</b> + glutamine). The reciprocal ATP/GTP requirement keeps the two pools balanced. Atoms come from glycine, glutamine, aspartate, CO <sub>2</sub> and formyl-THF. |
| <b>Purine —<br/>salvage</b>       | <b>HGPRT</b> (hypoxanthine, guanine) and <b>APRT</b> (adenine) reattach free bases to PRPP — energetically far cheaper, and vital in brain and bone marrow, which cannot synthesise de novo.                                                                                                                                                                                                                                                 |
| <b>Purine —<br/>catabolism</b>    | AMP/GMP → inosine/guanosine → <b>hypoxanthine</b> → <b>xanthine</b> → <b>uric acid</b> by <b>xanthine oxidase (a molybdenum flavoprotein of liver and intestinal mucosa)</b> . Most mammals then convert uric acid to soluble <b>allantoin</b> by <b>uricase</b> .                                                                                                                                                                           |
| <b>Pyrimidine</b>                 | The <b>ring is made first and attached to PRPP afterwards</b> . <b>CPS-II</b> (cytosolic, uses glutamine, <b>needs no N-acetylglutamate</b> ) is rate-limiting → carbamoyl aspartate → dihydroorotate → orotate → OMP → <b>UMP</b> → <b>UTP</b> → <b>CTP</b> . <b>dTMP</b> comes from dUMP by <b>thymidylate synthase</b> using N <sup>5</sup> ,N <sup>10</sup> -methylene-THF.                                                              |
| <b>Deoxy-<br/>ribonucleotides</b> | <b>Ribonucleotide reductase</b> reduces NDP → dNDP using <b>thioredoxin</b> ; allosterically regulated; inhibited by hydroxyurea.                                                                                                                                                                                                                                                                                                            |

**The three points that always appear:** ① **Birds and reptiles are uricotelic** — they lack uricase and excrete uric acid as a paste; overload gives **avian visceral and articular gout** (high-protein ration, renal damage, water deprivation). ② The **Dalmatian dog** handles urate abnormally and forms **urate uroliths**. ③ **Allopurinol** is a **suicide inhibitor of xanthine oxidase**; **5-fluorouracil** blocks thymidylate synthase and **methotrexate** blocks dihydrofolate reductase — the classical antimetabolites. Pyrimidine catabolism, by contrast, gives soluble β-alanine and β-aminoisobutyrate, so **pyrimidines never form stones**.

## 19 DNA & RNA BIOSYNTHESIS AND ITS REGULATION

**Replication** is **semi-conservative** (Meselson–Stahl), bidirectional from an origin, and every polymerase works **only 5'→3'**.

Helicase unwinds → SSB + topoisomerase/gyrase → Primase lays RNA primer →  
DNA pol III extends → Pol I removes primer → Ligase seals

- **Leading strand continuous; lagging strand discontinuous as Okazaki fragments.** Fidelity comes from the **3'→5' exonuclease proofreading** of the polymerase. Eukaryotic polymerases:  $\alpha$  (primase),  $\delta$  and  $\epsilon$  (elongation),  $\gamma$  (mitochondrial); telomerase extends chromosome ends.
- **Transcription:** DNA-dependent RNA polymerase, **needs no primer**, reads the **template (antisense) strand 3'→5'** and builds RNA 5'→3'. Prokaryotic core  $\alpha_2\beta\beta'\omega$  + **sigma factor** for promoter recognition (**Pribnow box TATAAT at -10**), **rho factor** for termination. Eukaryotes: **RNA pol I → rRNA, pol II → mRNA, pol III → tRNA & 5S**; TATA (Hogness) box.
- **Post-transcriptional processing** of eukaryotic mRNA: **5' capping with 7-methylguanosine, 3' polyadenylation, and splicing out of introns by the spliceosome.**
- **Inhibitors worth naming:** **rifampicin** → bacterial RNA polymerase; **actinomycin D** → intercalates DNA;  **$\alpha$ -amanitin** → RNA pol II (**Amanita** mushroom poisoning); **fluoroquinolones** → DNA gyrase; acyclovir/AZT → nucleoside analogues.
- **Regulation — Jacob & Monod operon model. lac operon (inducible):** allolactose inactivates the repressor, transcription begins; superimposed **catabolite repression** by glucose via cAMP–CAP. **trp operon (repressible):** tryptophan acts as co-repressor and switches the operon off. In eukaryotes: promoters and **enhancers, transcription factors, steroid-hormone response elements, DNA methylation, histone acetylation and miRNA.**
- **Translation:** the genetic code is a **triplet, degenerate, non-overlapping, comma-less and near-universal**; **AUG** starts, **UAA/UAG/UGA** stop; wobble at the third base. Antibiotics: **streptomycin & aminoglycosides (30 S, misreading), tetracycline (30 S A-site), chloramphenicol (50 S peptidyl transferase), macrolides (50 S translocation).**

## 20 REGULATION & INTEGRATION OF METABOLISM

**Three junction molecules** decide the fate of everything: **glucose-6-phosphate** (glycogen / glycolysis / HMP shunt), **pyruvate** (lactate / acetyl-CoA / OAA / alanine) and **acetyl-CoA** — the true crossroads, from which carbon can go to **TCA, fatty acids, cholesterol or ketone bodies** but **never back to glucose**.

| ORGAN                  | METABOLIC CHARACTER                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Liver</b>           | The <b>glucostat</b> and central processing plant — glycogen, gluconeogenesis, ketogenesis, urea, lipoproteins, detoxification |
| <b>Brain</b>           | Obligate glucose user; switches to <b>ketone bodies</b> only after days of starvation; no fuel stores                          |
| <b>Skeletal muscle</b> | Glycogen <b>for its own use only</b> ; the body's <b>protein reserve</b> ; exports alanine and lactate                         |
| <b>Adipose tissue</b>  | Triacylglycerol store; hormone-sensitive lipase is the tap                                                                     |
| <b>RBC</b>             | <b>Anaerobic glycolysis only</b> — no mitochondria                                                                             |
| <b>Kidney</b>          | Gluconeogenesis and ammoniogenesis in acidosis                                                                                 |

**Fed state (insulin dominant):** glucose uptake via GLUT-4, **glycogenesis, lipogenesis, protein synthesis**; glycolysis and the HMP shunt run; PDH is dephosphorylated and active.

**Fasting and starvation (glucagon, adrenaline, cortisol, GH):** a strict sequence — ① **hepatic glycogenolysis** (first 12–24 h) → ② **gluconeogenesis** from amino acids, lactate and glycerol → ③ **lipolysis and ketogenesis**, which spare glucose for the brain → ④ **protein catabolism** as the last resort. Muscle switches to fatty acids and ketones within days.

| HORMONE             | EFFECT ON BLOOD GLUCOSE                                                   |
|---------------------|---------------------------------------------------------------------------|
| Insulin             | <b>The only hypoglycaemic hormone</b> — uptake, glycogenesis, lipogenesis |
| Glucagon            | ↑ glycogenolysis, gluconeogenesis, ketogenesis (liver only)               |
| Adrenaline          | ↑ glycogenolysis in liver <b>and</b> muscle; lipolysis                    |
| Cortisol            | ↑ gluconeogenesis, protein catabolism, insulin resistance                 |
| GH, thyroxine, ACTH | Hyperglycaemic, anti-insulin                                              |

**Ruminant integration — write this as your conclusion in any long answer:** the ruminant absorbs virtually no glucose. Instead it absorbs **VFAs: acetate (≈65 %) → lipogenesis and milk fat; propionate (≈20 %) → the sole major gluconeogenic precursor; butyrate (≈15 %) → converted to β-hydroxybutyrate by the rumen wall**. Therefore the ruminant is permanently gluconeogenic, permanently mildly ketogenic, and any negative energy balance — **peak lactation, twin pregnancy, transport, anorexia** — tips it straight into **ketosis or pregnancy toxemia**. Everything in Unit 3's energy-disorder section is this paragraph applied clinically.

**Trap:** the ruminant's low blood glucose (45–60 mg/dL) is **normal**, not hypoglycaemia; and a "normal" glucose in a ketotic cow does not exclude ketosis — **BHB, not glucose, is the diagnostic analyte**.

## 21 FINAL RECALL BANK — READ THESE LAST, STRAIGHT BEFORE THE PAPER

|                                      |                                                      |
|--------------------------------------|------------------------------------------------------|
| Enzyme class of transaminases        | <b>Transferases (class 2)</b>                        |
| Enzyme class needing ATP to join     | <b>Ligases / synthetases (6)</b>                     |
| Apoenzyme + cofactor                 | <b>Holoenzyme</b>                                    |
| Coenzyme of transamination           | <b>Pyridoxal phosphate (B<sub>6</sub>)</b>           |
| Coenzyme of carboxylation            | <b>Biotin</b>                                        |
| Vitamin bound by avidin              | <b>Biotin</b>                                        |
| Coenzyme of methylmalonyl-CoA mutase | <b>Vitamin B<sub>12</sub> (cobalt)</b>               |
| Selenium enzyme                      | <b>Glutathione peroxidase</b>                        |
| Molybdenum enzyme                    | <b>Xanthine oxidase</b>                              |
| LDH isoenzyme of heart / liver       | <b>LDH-1 (H<sub>4</sub>) / LDH-5 (M<sub>4</sub>)</b> |

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| ALP isoenzyme unique to dogs            | <b>Corticosteroid-induced C-ALP</b>        |
| Modern concept of the active site       | <b>Koshland's induced fit</b>              |
| Enzyme with absolute specificity        | <b>Urease</b>                              |
| Enzyme showing stereospecificity        | <b>Fumarase (trans only)</b>               |
| K <sub>m</sub> equals                   | <b>[S] at ½ V<sub>max</sub></b>            |
| Low K <sub>m</sub> means                | <b>High substrate affinity</b>             |
| K <sub>m</sub> independent of           | <b>Enzyme concentration</b>                |
| Lineweaver–Burk x-intercept             | <b>-1/K<sub>m</sub></b>                    |
| Definition of 1 katal                   | <b>1 mol substrate per second</b>          |
| 1 katal in IU                           | <b>6 × 10<sup>7</sup> IU</b>               |
| Index of enzyme purity                  | <b>Specific activity</b>                   |
| Highest turnover number                 | <b>Carbonic anhydrase</b>                  |
| Malonate inhibits                       | <b>Succinate DH — competitive</b>          |
| Cyanide inhibits                        | <b>Cytochrome oxidase (Complex IV)</b>     |
| Organophosphates inhibit                | <b>Acetylcholinesterase — irreversible</b> |
| Inhibition with parallel LB lines       | <b>Uncompetitive</b>                       |
| Kinetics of allosteric enzymes          | <b>Sigmoid, not Michaelis–Menten</b>       |
| Chemiosmotic theory given by            | <b>Peter Mitchell, 1961</b>                |
| Complex that pumps no protons           | <b>Complex II (SDH)</b>                    |
| Inhibitor of ATP synthase               | <b>Oligomycin</b>                          |
| Classical uncoupler                     | <b>2,4-Dinitrophenol</b>                   |
| Physiological uncoupler                 | <b>Thermogenin (UCP-1), brown fat</b>      |
| P : O ratio of NADH / FADH <sub>2</sub> | <b>2.5 / 1.5</b>                           |

|                                             |                                                  |
|---------------------------------------------|--------------------------------------------------|
| Rate-limiting enzyme of glycolysis          | <b>Phosphofructokinase-1</b>                     |
| Glycolysis inhibitor used in glucose tubes  | <b>Sodium fluoride (enolase)</b>                 |
| Liver glucose sensor enzyme                 | <b>Glucokinase (high K<sub>m</sub>)</b>          |
| Net ATP of anaerobic glycolysis             | <b>2</b>                                         |
| Coenzymes of the PDH complex                | <b>TPP, lipoamide, CoA, FAD, NAD<sup>+</sup></b> |
| Rate-limiting enzyme of TCA                 | <b>Isocitrate dehydrogenase</b>                  |
| Only substrate-level phosphorylation in TCA | <b>Succinyl-CoA synthetase (GTP)</b>             |
| Fluoroacetate poisoning blocks              | <b>Aconitase</b>                                 |

|                                                |                                           |
|------------------------------------------------|-------------------------------------------|
| ATP per acetyl-CoA in TCA                      | <b>10</b>                                 |
| Total ATP per glucose                          | <b>32 (30 via glycerol-P shuttle)</b>     |
| Rate-limiting enzyme of HMP shunt              | <b>Glucose-6-P dehydrogenase</b>          |
| Two products of the HMP shunt                  | <b>NADPH + ribose-5-phosphate</b>         |
| Enzyme absent in muscle                        | <b>Glucose-6-phosphatase</b>              |
| Chief gluconeogenic substrate in ruminants     | <b>Propionate</b>                         |
| ATP cost of gluconeogenesis                    | <b>6 per glucose</b>                      |
| Muscle lactate → liver glucose                 | <b>Cori cycle</b>                         |
| Glycogen synthesis donor                       | <b>UDP-glucose</b>                        |
| Glycogen primer protein                        | <b>Glycogenin</b>                         |
| Coenzyme of glycogen phosphorylase             | <b>Pyridoxal phosphate</b>                |
| Rate-limiting step of $\beta$ -oxidation entry | <b>Carnitine palmitoyl transferase-I</b>  |
| CPT-I is inhibited by                          | <b>Malonyl-CoA</b>                        |
| ATP from complete oxidation of palmitate       | <b>106</b>                                |
| Rate-limiting enzyme of ketogenesis            | <b>HMG-CoA synthase</b>                   |
| Major ketone body                              | <b><math>\beta</math>-hydroxybutyrate</b> |
| Enzyme the liver lacks for ketone use          | <b>Thiophorase</b>                        |
| Rate-limiting enzyme of lipogenesis            | <b>Acetyl-CoA carboxylase (biotin)</b>    |
| Fatty acid synthase stops at                   | <b>Palmitate, C16</b>                     |
| NADPH needed per palmitate                     | <b>14</b>                                 |
| Chief oxidative deamination enzyme             | <b>Glutamate dehydrogenase</b>            |
| Chief transporter of ammonia                   | <b>Glutamine</b>                          |
| Rate-limiting enzyme of urea cycle             | <b>CPS-I (needs N-acetylglutamate)</b>    |
| ATP spent per molecule of urea                 | <b>4 high-energy equivalents</b>          |
| Liver-specific urea cycle enzyme               | <b>Arginase (<math>Mn^{2+}</math>)</b>    |
| Committed step of purine synthesis             | <b>PRPP amidotransferase</b>              |
| Enzyme inhibited by allopurinol                | <b>Xanthine oxidase</b>                   |
| Operon model proposed by                       | <b>Jacob &amp; Monod</b>                  |

**Thirty-second closing sweep:** OTHLIL six classes ·  $K_m = [S]$  at  $\frac{1}{2}V_{max}$ ; competitive raises  $K_m$ , non-competitive lowers  $V_{max}$  · 1 katal =  $6 \times 10^7$  IU · malonate  $\perp$  SDH, cyanide  $\perp$  complex IV, oligomycin  $\perp$  complex V, DNP uncouples · PFK-1 in, F-1,6-BPase out · isocitrate DH is TCA's rate-limiter; 10 ATP per acetyl-CoA · G-6-PD makes NADPH; onion & paracetamol break the shield · CPT-I blocked by malonyl-CoA; palmitate = 106 ATP · HMG-CoA synthase makes ketones, liver lacks thiophorase · CPS-I needs N-acetylglutamate; 4 ~P per urea · acetyl-CoA can never become glucose · propionate is the ruminant's glucose.